

Online Appendix to “Monopsony Makes it Big”

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A. Theoretical Appendix

A.1 Microfoundations for Firm-Specific Labor Supply Elasticity

Recall that worker i 's (unconditional) labor supply to firm j —the probability that i chooses j , integrated over ϵ_{ij} and $\{\epsilon_{ik}\}_{k \neq j}$ —is given by:

$$P[i \text{ chooses } j | w_j, \mathbf{w}_{-j}, S_i] = P \left(j = \left[\arg \max_{k \in S_i} V_i^k(w_k) \right] \right) \quad (\text{A.1})$$

where $\mathbf{w}_{-j}$ is a vector of wages offered by firms other than j . We use $k^* := \arg \max_{k \in S_i} V_i^k(w_k)$ to denote the best job for i .

Conditioning on the j -specific independent taste shock for i , ϵ_{ij} , in this

expression allows Equation A.1 to be expressed as a product of pairwise choice probabilities, including non-employment:¹

$$\begin{aligned} P[\text{i chooses } j \mid w_j, \mathbf{w}_{-j}, \epsilon_{ij}, S_i] &= \underbrace{P(V_i^j(w_j) > V_i^0 \mid w_j, \epsilon_{ij}, b)}_{\text{prob. } j \text{ better than unemp}} \\ &\times \underbrace{\prod_{k \in S_i \setminus \{j \cup o\}} P(V_i^j(w_j) > V_i^k(w_k) \mid w_j, \mathbf{w}_{-j}, \epsilon_{ij}, S_i)}_{\text{pairwise choice probabilities}} \end{aligned}$$

Log-differentiating the conditional probability above yields a conditional elasticity, which is an intermediate object—not the labor supply elasticity itself:

$$\eta_{ij}(w_j \mid S_i, b_i, \epsilon_{ij}) = \underbrace{\sum_{k \neq j, k > 0} \eta_{ijk}(w_j \mid S_i, b_i, \epsilon_{ij})}_{\text{Competition with other firms in } S_i} + \underbrace{\eta_{ij0}(w_j \mid S_i, b_i, \epsilon_{ij})}_{\text{Extensive margin from non-employment}}$$

This conditional elasticity is a stepping stone in the calculations, not the labor supply elasticity. To recover worker i 's labor supply to firm j , we first integrate over ϵ_{ij} (holding S_i fixed):

$$P[\text{i chooses } j \mid w_j, \mathbf{w}_{-j}, b_i, S_i] = \int P[\text{i chooses } j \mid w_j, \mathbf{w}_{-j}, \epsilon_{ij}, S_i, b_i] dF_\epsilon(\epsilon_{ij} \mid i) \quad (\text{A.2})$$

Consideration elasticity: We now incorporate endogenous S_i into the residual labor-supply elasticity decomposition. In general, modeling how a set of options responds to the price of a single one is a complicated problem. To

¹The independence of the probability of choosing the other jobs or the outside option requires conditioning on the j -specific taste shock ϵ_{ij} . Integrating over this shock would lead to dependence. The conditioning here is purely calculational—both ϵ_{ij} and $\{\epsilon_{ik}\}_{k \neq j}$ are random variables. Integrating back over ϵ_{ij} , as in Equation A.2, recovers worker i 's unconditional labor supply.

illustrate our main points, we take a much-simplified approach, and assume an own elasticity, leaving out microfoundations due to directed search, for example. We assume that firm j is in the consideration set S_i with probability $\pi_{ij}(w_j)$, which has an elasticity

$$\eta_{ij}^{\pi} := \frac{w_j}{\pi_{ij}} \frac{d\pi_{ij}}{dw_j}. \quad (\text{A.3})$$

Note that we assume that π_{ij} only depends on w_j (no cross partials, and no dependence on ϵ_{ij}), so that consideration sets are independent of both other firms' wages and idiosyncratic tastes.

Combining this with A.2, we can get a complete expression for the worker-specific supply to firm j as:

$$s_{ij} = \int \pi_{ij}(w_j) \mathbb{P}[\text{i chooses } j \mid w_j, \mathbf{w}_{-j}, \epsilon_{ij}, S_i, b_i] dF_{\epsilon}(\epsilon_{ij} | i) \quad (\text{A.4})$$

The corresponding worker-specific elasticity, still an intermediate object, is:

$$\eta_{ij}(w_j | b_i, \epsilon_{ij}) = \underbrace{\eta_{ij}^{\pi}}_{\text{Consideration}} + \underbrace{\sum_{k \neq j, k > 0} \eta_{ijk}(w_j | S_i, b_i, \epsilon_{ij})}_{\text{Competition with other firms in } S_i} + \underbrace{\eta_{ij0}(w_j | S_i, b_i, \epsilon_{ij})}_{\text{Extensive margin from non-employment}}$$

For each worker i we now have a elasticity that depends on tastes, the distribution of consideration sets, as well as outside options, that we can aggregate to get the labor supply facing the firm. Integrating over the distribution of workers i , we finally get the overall labor supply facing the firm as:

$$s_j = \int \int \pi_{ij}(w_j) P[\mathbf{i} \text{ chooses } \mathbf{j} \mid w_j, \mathbf{w}_{-\mathbf{j}}, \epsilon_{ij}, S_i, b_i] dF_\epsilon(\epsilon_{ij} \mid i) dF(i) \quad (\text{A.5})$$

We can then get a general expression for the elasticity to firm j

$$\eta_j = \frac{w_j}{s_j} \frac{ds_j}{dw_j} = \tilde{E}_i[\eta_{ij}^\pi] + \sum_{k \neq j, k > 0} \tilde{E}_i[\eta_{jk}] + \tilde{E}_i[\eta_{j0}] \quad (\text{A.6})$$

where $\tilde{E}_i[\mathbf{f}] \equiv \int \int \frac{P_{ij}}{s_j} \pi_{ij} \mathbf{f} dF_\epsilon(\epsilon_{ij} \mid i) dF(i)$.

While this expression is very general, and summarizes the three main forces driving monopsony, it is not clear how one would operationalize this in a parametric model. For this reason, we now specialize to the common log utility, logit error specification in the next section.

Logit-Log Specialization

Assume $\epsilon_{ik} \sim \text{iid Gumbel}(0, \sigma_i)$ and $U_k(w_k) = \beta \log w_k$ for all firms k , and $U_o(b_i) = \beta \log(b_i)$ for non-employment. Integrating (A.2) over ϵ_{ij} yields the multinomial logit:

$$s_{ij} := P[\mathbf{i} \text{ chooses } \mathbf{j} \mid w_j, \mathbf{w}_{-\mathbf{j}}, b_i, S_i] = \frac{w_j^{\beta/\sigma_i}}{b_i^{\frac{\beta}{\sigma_i}} + \sum_{k \in S_i \setminus \{o\}} w_k^{\beta/\sigma_i}}. \quad (\text{A.7})$$

Define the outside-option share analogously:

$$s_{i0} := \frac{b_i^{\frac{\beta}{\sigma_i}}}{b_i^{\frac{\beta}{\sigma_i}} + \sum_{k \in S_i \setminus \{o\}} w_k^{\beta/\sigma_i}}. \quad (\text{A.8})$$

Write $\alpha_i := \beta/\sigma_i$ (worker-specific β_i can be accommodated without loss).

Derivative of the conditional choice probability

Holding the consideration set constant, and differentiating (A.7) with respect to w_j gives:

$$\frac{ds_{ij}}{dw_j} = s_{ij} (1 - s_{ij}) \frac{\alpha_i}{w_j} = s_{ij} \left(s_{i0} + \sum_{k \neq j, k > 0} s_{ik} \right) \frac{\alpha_i}{w_j}, \quad (\text{A.9})$$

where the second equality uses $1 - s_{ij} = s_{i0} + \sum_{k \neq j, k > 0} s_{ik}$.

Aggregate share and its elasticity

The share of workers at firm j is given by:

$$s_j = \int \pi_{ij}(w_j) s_{ij}(w_j) dF(i), \quad (\text{A.10})$$

where $dF(i)$ integrates over worker heterogeneity (including consideration probabilities π_{ik} as well as outside options b_i or preference heterogeneity σ_i).

Differentiating and multiplying by w_j/s_j we get the own-wage labor supply elasticity to firm j :

$$\begin{aligned} \eta_j &= \frac{w_j}{s_j} \frac{ds_j}{dw_j} \\ &= \frac{w_j}{s_j} \int \left[\frac{d\pi_{ij}}{dw_j} s_{ij} + \pi_{ij} \frac{ds_{ij}}{dw_j} \right] dF(i) \\ &= \frac{1}{s_j} \int \pi_{ij} s_{ij} \left[\underbrace{\frac{w_j}{\pi_{ij}} \frac{d\pi_{ij}}{dw_j}}_{\eta_{ij}^\pi} + \underbrace{(1 - s_{ij}) \alpha_i}_{\text{from (A.9)}} \right] dF(i). \end{aligned} \quad (\text{A.11})$$

Expanding $1 - s_{ij} = s_{i0} + \sum_{k \neq j, k > 0} s_{ik}$:

$$\eta_j = \frac{1}{s_j} \int \pi_{ij} s_{ij} \left[\underbrace{\eta_{ij}^\pi}_{\text{consideration}} + \underbrace{\alpha_i \sum_{k \neq j, k > 0} s_{ik}}_{\text{competition across firms}} + \underbrace{\alpha_i s_{i0}}_{\text{exit to outside}} \right] dF(i). \quad (\text{A.12})$$

Three-part decomposition

Define the component elasticities:

$$\eta_j^{\text{consideration}} = \frac{1}{s_j} \int \pi_{ij} s_{ij} \eta_{ij}^\pi dF(i), \quad (\text{A.13})$$

$$\sum_{k \neq j} \eta_{jk} = \frac{1}{s_j} \int \pi_{ij} s_{ij} \left(\sum_{k \neq j, k > 0} s_{ik} \right) \alpha_i dF(i). \quad (\text{A.14})$$

$$\eta_{j0} = \frac{1}{s_j} \int \pi_{ij} s_{ij} s_{i0} \alpha_i dF(i), \quad (\text{A.15})$$

So the overall labor supply elasticity to firm j can be written as:

$$\eta_j = \eta_j^{\text{consideration}} + \sum_{k \neq j, k > 0} \eta_{jk} + \eta_{j0}. \quad (\text{A.16})$$

Interpretation

- **Consideration margin** ($\eta_j^{\text{consideration}}$): A higher wage draws more workers into considering firm j . This operates before any preference comparison.
- **Competition Across Firms** ($\sum_{k \neq j, k > 0} \eta_{jk}$): A higher wage poaches workers from competing firms k . This is the intensive (job-switching) mar-

gin.

- **Exit to Non-employment** (η_{j0}): A higher wage pulls workers from nonemployment into firm j . This is the extensive (participation) margin.

A.2 Dynamic search as a limit of discrete choice

While our model incorporates the idea that workers' consideration sets may respond to wages, it is more common in the monopsony literature to invoke random search. Random search models focus on the rate at which workers obtain offers from the firm distribution of wages $F(w)$ (at rate λ), and the rate at which they exogenously lose jobs (rate δ). Denote the share unemployed by $u(t)$.

The connection to our framework is direct: in a dynamic search model, an employed worker's effective consideration set at any instant consists of their current employer together with—with probability $\lambda\Delta t$ in a small interval Δt —a single randomly drawn outside firm. The choice between the current and the outside firm is deterministic when there is no preference heterogeneity: the worker takes the higher-paying job.

Writing this in our notation, the dynamic labor supply facing a single firm is given by:

$$\begin{aligned} P(j|w_j, \mathbf{w}_{-j}, t + \Delta t) &= (1 - \Delta t\delta)P(j, t) \\ &+ \Delta t\lambda \left(\sum_{k \neq j} [P(k, t)f(j)P(k \rightarrow j|w_j, \mathbf{w}_{-j}) - P(j, t)f(k)P(j \rightarrow k|w_j, \mathbf{w}_{-j})] \right) \\ &+ \Delta t\lambda u(t)f(j) \end{aligned}$$

where $P(k, t)$ denotes the share of the total workforce at firm k at time t (so that $\sum_k P(k, t) + u(t) = 1$), and $f(j)$ is the probability that a random offer is drawn from firm j . The term $f(j)$ ensures that only offers specifically from firm j contribute to its recruitment (and similarly $f(k)$ for the separation flow to firm k). Note that

no additional factor of $(1 - u(t))$ appears in the job-to-job flow terms because the employment level is already captured in $P(k, t)$. For simplicity, let all workers have a common reservation wage b . Because the probability that a worker transitions from k to j is a step function:

$$P(k \rightarrow j | w_j, \mathbf{w}_{-j}) = \mathbf{1}_{w_k < w_j},$$

the cross-firm elasticity is effectively zero, except in the zero probability case of $w_k = w_j$. However, firms face upward-sloping supply curves because there is an asymmetry between recruits and separations: recruits are pulled from the firms below j in the job ladder while separations are lost to firms above j in the job ladder. By toggling the wage, an employer changes their rank in the wage distribution, altering the relative flows of recruits and separations, which changes the labor supply to the firm.

We can obtain this labor supply elasticity by taking limits in Δt and approximating sums by integrals. Letting $p(w, t)$ denote per-firm employment (i.e., the number of workers at a firm offering wage w) at time t , this gives us the following equation for the evolution of $p(w, t)$ from Burdett and Mortensen (1998):

$$\frac{dp(w, t)}{dt} = \lambda(G(w, t) - p(w, t)(1 - F(w))) - \delta p(w, t)$$

while unemployment evolves as:

$$\frac{du(t)}{dt} = \delta(1 - u(t)) - \lambda u(t)$$

Here $G(w, t)$ denotes the total pool of workers who would accept an offer at wage w :

$$G(w, t) \equiv u(t) + \int_b^w f(w')p(w', t) dw'$$

which comprises all unemployed workers (who accept any offer) plus the

mass of employed workers currently earning below w . Note that $f(w')p(w', t)$ gives the density of employed workers across wages, since $p(w', t)$ is per-firm employment and $f(w')$ is the density of wage offers.

The steady-state firm-level employment is given by:

$$p(w) = \frac{\lambda G(w)}{\lambda(1 - F(w)) + \delta}$$

where we write $G(w)$ as the steady-state $G(w, t)$. Unemployment is given by:

$$u = \frac{\delta}{\delta + \lambda}$$

G and p are jointly determined in steady state (since $G(w)$ depends on $p(w')$ for $w' < w$, and p depends on G). This joint determination yields per-firm employment levels that are in general nonlinear functions of (log) wages, and hence yield non-constant elasticities of labor supply facing the firm. Note that the flow of separations is $s(w) = \lambda(1 - F(w)) + \delta$, while the recruit inflow is $\lambda((1 - u)G(w) + u)$. In steady state, these flows are equal, which yields the steady-state firm-level labor supply, $p(w)$. This also highlights an important aspect of the residual labor supply elasticity: that it is the sum of the recruit and (negative) separation elasticities.

The steady-state labor supply elasticity. Log-differentiating the steady-state expression $p(w) = \frac{\lambda(u+(1-u)G(w))}{s(w)}$ with respect to $\log w$, where $s(w) = \lambda(1 - F(w)) + \delta$, yields:

$$\eta(w) \equiv \frac{d \log p(w)}{d \log w} = \underbrace{\frac{(1-u)wG'(w)}{u + (1-u)G(w)}}_{\eta^R(w)} + \underbrace{\frac{w\lambda f(w)}{s(w)}}_{-\eta^S(w)} \quad (\text{A.17})$$

The first term is the recruit elasticity $\eta^R(w)$ —how the recruitable pool responds to the wage—and the second is $-\eta^S(w)$, the negative of the separa-

tion elasticity, since a higher wage reduces the separation rate by moving the firm up the offer distribution. Both components are wage-dependent. Since in steady state $G'(w) = f(w)p(w)$ and $p(w) = \lambda(u + (1 - u)G(w))/s(w)$, the recruit elasticity can be written as $\eta^R(w) = \frac{wf(w)p(w)}{u + (1 - u)G(w)} = \frac{\lambda wf(w)}{s(w)}$, which implies $\eta^R(w) = -\eta^S(w)$ —the doubling relationship holds at every point in the wage distribution (not just at a symmetric equilibrium), because it is a consequence of the steady-state flow balance. The total elasticity is therefore:

$$\eta(w) = \frac{2\lambda wf(w)}{s(w)} = \frac{2\lambda wf(w)}{\lambda(1 - F(w)) + \delta}$$

This expression shows clearly how monopsony power varies across the wage distribution. At the bottom of the distribution (near the reservation wage b), $s(w)$ is large (many firms offer more, so separations are frequent) and $f(w)$ may be small, yielding a low elasticity and substantial monopsony power. At the top, $s(w) \approx \delta$ (only exogenous separations remain), so the elasticity can be very high. This illustrates how a single structural model can generate wide variation in monopsony power across the wage distribution.

Many empirical papers in the monopsony literature estimate the amount of monopsony power held by firms by estimating the wage elasticity of separations and using this to calculate the total labor supply elasticity to the firm (comprising both the elasticity of separations and recruitment). This is because there is a steady state relationship between the separation and recruit elasticities—namely that $\eta^R = -\eta^S$ —which holds in a broad class of dynamic models, provided that there is no wage impact on transitions to unemployment (Manning, 2003).² where the elasticities are evaluated at the equilibrium values. The key property ensuring $\eta^R = -\eta^S$ is steady-state bal-

²To see this, note the probability of transitioning from firm j to firm k is equal to $P(k|j, \mathbf{w})$ where $\mathbf{w}$ is a vector of wages. Recruits are given by $R_j = \sum_{k \neq j} P(j|k, \mathbf{w})P(k)N$ and separations are given by $(1 - P(j|j, \mathbf{w})) (P(j)N) = s_j N_j$. In steady state, the number of workers joining a firm (from all firms) must equal the number leaving: $R_j = s_j N_j$. Log differentiating this expression yields the familiar expression: $\eta_j^N = \eta_j^R - \eta_j^S$

ance, so the marginal effect of firm j raising its wage on its recruit inflow (through the transition probability $Pr(k \rightarrow j|w_j, w_k)$) is the mirror image of the effect on its quit outflow (through $Pr(j \rightarrow k|w_j, w_k)$). As shown above, in the Burdett-Mortensen model this relationship holds throughout the wage distribution as a consequence of steady-state flow balance. If each elasticity is a constant, this implies that the recruit and separation elasticities are equal in absolute value, and that the overall labor supply elasticity can be calculated by multiplying the separation elasticity by -2 .

Search with random utility. Generally, however, search models do not lead to constant elasticity labor supply functions. One way to get a constant elasticity labor supply function in a search model is by combining the search structure of consideration sets with random utility draws, as in Bloesch et al. (2026), albeit with a degenerate endogenous wage distribution. Suppose, as in standard search, employed workers only have consideration sets of at most two firms—their current employer and one outside offer. Adding type-1 extreme value preference shocks with scale parameter $1/\eta$ to log-wage utility, the probabilities of a worker transitioning from k to j (a recruitment event for firm j) and from j to k (a separation from firm j) are:

$$P(k \rightarrow j|w_j, \mathbf{w}_{-j}) = \frac{w_j^\eta}{w_j^\eta + w_k^\eta} \quad \text{and} \quad P(j \rightarrow k|w_j, \mathbf{w}_{-j}) = \frac{w_k^\eta}{w_j^\eta + w_k^\eta}$$

where the logistic form follows from the standard result for pairwise choice under type-1 extreme value shocks.

In their model, Bloesch et al. (2026) show that, integrating over the distribution of vacancies offered and worker wages, these pairwise consideration sets generate a constant labor supply elasticity, equal to the sum of the recruitment elasticity and the (negative of) the separation elasticity. We refer

readers to their paper for further details.

A.3 Static Monergony

In this appendix we present the full monergony model introduced in Section 4—a combination of efficiency wages and monopsony in which both the effort elasticity and the residual labor supply elasticity to the firm are finite.

We illustrate the role of efficiency-wage considerations in a monopsony setting using a labor-discipline model (Shapiro and Stiglitz, 1984; Bowles, 1985). If effort is non-contractible and hard to monitor, a higher wage raises the cost of job loss and thereby reduces shirking. We show how the distribution of outside options shapes wage setting even when effort elicitation matters. When the firm can screen on outside options, wage setting is governed by the elasticity of the *effective* labor supply of non-shirkers, and workers receive an “efficiency wage premium” above the classic monopsony wage to deter shirking. Here the firm always exercises monopsony power and is “labor-hungry,” but there is also involuntary rationing: some workers willing to take the job at the posted wage are turned away because they would shirk. When screening is not available, some workers shirk in equilibrium. The firm may be (i) supply-constrained, when both effort elicitation and participation matter, leading to a markdown, or (ii) demand-constrained, when effort extraction is sufficiently costly and the optimal wage is sufficiently high that the supply constraint does not bind, and jobs are randomly rationed. Even then, the Solow wage depends on the shape of the outside-option distribution.

Model setup: Workers have heterogeneous outside options b_i with smooth CDF $F(\cdot)$ and density $f(\cdot)$. In standard monopsony, the firm posts w and all with $b_i \leq w$ work and produce the same output. In our setting, each worker sees w and her private b_i and chooses effort $e_i \in \{0, 1\}$. The firm may be able

to detect b_i (depending on the full screening versus no-screening case); if it knows b_i it can condition hiring on b_i , but cannot wage discriminate. Importantly, it cannot observe effort, but uses a monitoring technology that detects shirking ($e_i = 0$) with probability $P \in (0, 1)$. A detected shirker is fired and receives b_i , so there is limited liability. A non-shirker produces one unit of effective labor at cost c .

Given the monitoring technology, a risk-neutral worker with outside option b_i exerts effort only if

$$\underbrace{w - c}_{\text{working}} \geq \underbrace{(1 - P) \times w}_{\text{not caught shirking}} + \underbrace{P \times b_i}_{\text{caught shirking}}$$

which can be rewritten as:

$$w \geq \frac{c}{P} + b_i \quad (\text{A.18})$$

The worker only accepts the offer if individual rationality (participation constraint) holds (i.e., if accepting the offer is weakly better than taking the outside option b_i):

$$\max\{\underbrace{w - c}_{\text{work}}, \underbrace{(1 - P) \times w + P \times b_i}_{\text{shirk}}\} \geq b_i$$

Since $P < 1$, if the incentive compatibility constraint is satisfied, the participation constraint is also satisfied. Let $\phi = \frac{c}{P}$ be the no-shirking premium that solves the no-shirking constraint for a worker, so that:

$$Pw + (1 - P)b_i - c \geq b_i \implies w \geq b_i + \phi$$

We use L to denote the total labor hired by the firm. For a firm offering w , the total supply of effective labor (which we denote $E(w)$) depends on both

whether workers accept the offer and whether they do not shirk (shirkers contribute nothing to production). Note that $E(w)$ denotes the maximum effective labor available at wage w ; realized effective labor may be lower if the firm is demand constrained. Further, for notational simplicity we do not condition all expressions on F .

The total supply of effective labor, assuming a unit mass of labor, is given by:

$$E(w) = \int \mathbf{1}_{w \geq b_i + \phi} dF(b_i) = F(w - \phi) \quad (\text{A.19})$$

The average effective labor per worker hired depends on whether the firm can screen on b_i :

$$e(w) = \begin{cases} \frac{F(w-\phi)}{F(w)} & \text{if screening on } b_i \text{ infeasible } (L \leq F(w)) \\ 1 & \text{if screening on } b_i \text{ feasible } (L \leq F(w - \phi)) \end{cases} \quad (\text{A.20})$$

With full screening, the firm rationally hires only workers with $b_i \leq w - \phi$ (non-shirkers). Without screening, only workers with $b_i \leq w - \phi$ provide effort, but workers with $b_i \leq w$ accept the job.

If $G(E)$ is a smooth, increasing, and concave production function, the firm's problem is either :

$$\max_{w,L} G(e(w)L) - wL \quad \text{s.t.} \quad L \leq F(w).$$

$$\max_{w,L} G(e(w)L) - wL \quad \text{s.t.} \quad L \leq F(w - \phi).$$

if screening is feasible. $e(w)$ is as defined in (A.20). In all cases, there are some parameterizations in which the firm would prefer to shut down; for simplicity, we assume the firm finds it profitable to operate.

Explicit Solution for Main Text Examples We next present explicit solutions for the three cases presented graphically in the main text. We deliberately chose tractable functional forms. In particular we assume a log production function: $G(e(w)L) = a \log(e(w)L)$. This ensures that the MRPL schedule $G'(e(w)L) = a/(e(w)L)$ depends on effective labor alone, so the MRPL curve does not shift across cases, facilitating a graphical presentation. Further, we suppose labor supply to the firm is a shifted-power function, as in Card et al. (2018), which can be microfounded as a random utility labor supply function with deterministic utility from a firm paying wage w_j given by $\kappa \log(w_j - \mu)$, where μ is a parameter of the labor supply function and workers have firm-specific idiosyncratic preferences (ϵ_{ij}) which follow a type-1 extreme value distribution. The resulting labor supply function to a firm is given by³:

$$L(w) = F(w) = \begin{cases} C(w - \mu)^\kappa, & \text{if } w \geq \mu, \\ 0, & \text{if } w < \mu, \end{cases} \quad \kappa > 0, (w - \mu)^\kappa \leq \frac{1}{C}$$

where $\frac{1}{C} = \sum_k (w_k - \mu)^\kappa$ is a normalization term that includes w_j and ensures that $F(w)$ is a distribution. Following Card et al. (2018) we assume the set of k firms is large so that the effect of the wage on the denominator C is negligible over the relevant range. Without loss of generality we set $C = 1$ and suppress it henceforth. The corresponding density of reservation utilities is

$$f(w) = F'(w) = \kappa(w - \mu)^{\kappa-1} \quad \text{for } w > \mu$$

³While these microfoundations are inconsistent with the risk-neutral worker utilities from the previous section, we merely need the shifted-power functional form for our illustrative calculations and closed form expressions. An alternative approach would be to directly assume the distribution of outside options have the shifted power form.

and the elasticity of labor supply is:

$$\eta(w) = \kappa \frac{w}{w - \mu}.$$

The log production function ensures independence of the marginal revenue product of labor from effort. The shifted-power labor supply function (used in Card et al. (2018)) makes the labor supply elasticity explicitly a function of μ . There is a simple comparison with Shapiro and Stiglitz (1984). It also nests the constant elasticity case, which is commonly used in the empirical and structural literature. A limitation of the constant elasticity case is that it does not allow the elasticity to be different at a wage $w > \mu$ versus at $w = \mu$, and as we discussed in the main text and show below, this obscures the relationship between the Shapiro-Stiglitz model and our model.

As discussed in the text, there are three relevant cases, which we summarize in Table A.1. For the figures in the text, we fix $a = .7$, $\kappa = 3$, $\mu = 0.2$, and $\phi = 0.08$ for Cases 1 and 2, and $\phi = 0.18$ for Case 3.

Table A.1: **Monergony Cases**

	Full Screening	No Screening
Labor supply constrained	Case 1: $e(w) = 1$ $L = F(w - \phi)$	Case 2: $e(w) = \frac{F(w - \phi)}{F(w)}$ $L = F(w)$
Labor demand constrained	—	Case 3: $e(w) = \frac{F(w - \phi)}{F(w)}$ $L < F(w)$

Note: When screening on b_i is possible (Case 1), only workers who will exert effort are hired, so there is (efficient) rationing of jobs. Without screening (Cases 2 and 3), some hired workers shirk, of whom a share P are fired. In Case 3, the firm hires a random subset of workers willing to work (random rationing).

Case 1: Full screening is feasible (labor supply binds in terms of effective labor). With full screening, the firm never hires workers who would shirk

(those with $b_i \in (w - \phi, w]$).⁴ Hence the total amount of labor equals the total amount of effective labor $L = E$. Employer screening leads jobs to be rationed (efficiently) as there are workers willing to work at the offered wage of w who are not hired because they have too high a b_i (as they would shirk). This is also a novel prediction of why employers may spend resources screening workers even as they are “labor hungry.”

The firm’s problem reduces to

$$\max_w \Pi(w) = G(E(w)) - wE(w), \quad E'(w) = f(w - \phi).$$

The first-order condition for the profit-maximizing wage, w_1 , is:

$$G'(E)E'(w_1) - (E(w_1) + w_1E'(w_1)) = 0 \iff G'(E) = \frac{E(w_1)}{E'(w_1)} + w_1 = \frac{F(w_1 - \phi)}{f(w_1 - \phi)} + w_1.$$

Rearranging,

$$w_1 = G'(E) - \frac{F(w_1 - \phi)}{f(w_1 - \phi)}. \quad (\text{A.21})$$

Let $\eta(x) \equiv \frac{x f(x)}{F(x)}$ denote the labor supply elasticity evaluated at x . Since $\frac{F(w - \phi)}{f(w - \phi)} = \frac{w - \phi}{\eta(w - \phi)}$, (A.21) implies

$$w_1 = G'(E) - \frac{w_1 - \phi}{\eta(w_1 - \phi)} \iff w_1 = \frac{\eta(w_1 - \phi)}{1 + \eta(w_1 - \phi)} \underbrace{G'(E)}_{\text{MRPL}} + \underbrace{\frac{\phi}{1 + \eta(w_1 - \phi)}}_{\text{Efficiency Wage Premium}} \quad (\text{A.22})$$

This is the standard monopsony markdown, but applied to *effective* labor E , plus an additional “Efficiency Wage Premium” term $\phi/(1 + \eta)$ arising from the no-shirking requirement. Intuitively, the firm exercises monopsony power over E (supply $F(w_1 - \phi)$), and all hires exert effort, so $G'(E)$ is

⁴The firm would increase profits by firing a shirker, since that would not affect total production, while lowering costs.

the MRPL of effective labor.

In Figure 4, this case can be shown as a shift back in the firm-specific labor supply, and hence the MCL curve, as compared to the classic monopsony case, because now a wage of w yields $E = F(w - \phi)$ instead of $F(w)$ workers. For a given number of effective workers, the MRPL is the same, $G'(E)$, because all hired workers exert effort when employers can perfectly screen based on outside options. Wages are higher than under classic monopsony. The full screening case provides a clear illustration of how incentive compatibility constraints can offset monopsony wage-setting.

We can also see why the bottom-left cell of Table A.1 is empty. Since employers can hire based on outside options, but cannot condition the wage on the outside option, they will always be (effective) labor supply constrained, and the traditional efficiency-wage condition, or Solow condition, will never bind. Suppose the number of hired workers (L) were less than the number of non-shirkers willing to work at the profit-maximizing wage w , so that $L < F(w - \phi)$. An employer who can screen could then lower the wage to w' such that $L = F(w' - \phi)$ and still hire L non-shirkers, thereby raising profits; this contradicts the claim that w was profit-maximizing. Furthermore, it never pays for a firm to hire more than $F(w - \phi)$ workers at wage w . If $L > F(w - \phi)$ at the optimum, some workers are shirking; a profit-maximizing employer with the ability to screen can fire an expected shirker, reducing costs without harming output. This contradicts the claim that L is profit-maximizing. Therefore, with full screening, profit maximization implies the effective labor supply constraint is always binding: $L = F(w - \phi)$.

Adding Functional Form Assumptions: With shifted-power labor supply, the binding supply constraint implies

$$L_1 = F(w_1 - \phi) = (w_1 - \phi - \mu)^\kappa, \quad e(w_1) = 1, \quad E_1 = L_1.$$

Profit as a function of w is

$$\begin{aligned}\Pi_1(w) &= G(E_1(w)) - wL_1(w) = a \ln \left((w - \phi - \mu)^\kappa \right) - w(w - \phi - \mu)^\kappa \\ &= a\kappa \ln(w - \phi - \mu) - w(w - \phi - \mu)^\kappa.\end{aligned}$$

The first-order condition $d\Pi_1/dw = 0$ is

$$\frac{a\kappa}{(w_1 - \phi - \mu)} - (w_1 - \phi - \mu)^\kappa - \kappa w_1 (w_1 - \phi - \mu)^{\kappa-1} = 0.$$

The marginal revenue product of labor at the optimum is

$$\text{MRPL}_1 = G'(E_1) = \frac{a}{E_1} = \frac{a}{(w_1 - \phi - \mu)^\kappa},$$

which is graphically illustrated with parameter values in Figure 4a.

Case 2: No screening, and labor supply binds. Now suppose the firm cannot condition hiring on μ , and the labor supply constraint binds. In this case effective labor is given by $E(w) = F(w - \phi)$, while employment is $L = F(w)$ as all workers with outside option worse than w will choose to work at the firm. The firm's problem is

$$\max_w G(F(w - \phi)) - wF(w).$$

Here, a higher wage raises effective labor through both participation and effort. Since $E'(w) = f(w - \phi)$, the first-order condition for w_2 (the optimal wage in the second case) is:

$$G'(E(w_2)) f(w_2 - \phi) - (F(w_2) + w_2 f(w_2)) = 0.$$

Note that the MCL term is exactly the same as in a monopsony problem,

while the MRPL is now modified by effort extraction. We can write the wage as a standard monopsony markdown applied to an effort-adjusted marginal revenue product of labor, MRPL_{eff} :

$$\text{MRPL}_{\text{eff}}(w_2) \equiv G'(F(w_2 - \phi)) \cdot \frac{f(w_2 - \phi)}{f(w_2)}, \quad w_2 = \frac{\eta(w_2)}{1 + \eta(w_2)} \text{MRPL}_{\text{eff}}(w_2).$$

The first factor, $G'(F(w - \phi))$, captures that, holding employment fixed, effective labor is scarcer when some workers shirk, so the marginal product per effective unit is higher. The second factor, $f(w_2 - \phi)/f(w_2)$, captures the *conversion rate* from a higher wage to additional effective labor through the no-shirking threshold. Relative to the classic monopsony case $E = F(w)$, these two forces can offset or reinforce each other, so the net shift in MRPL_{eff} (and hence in wages/employment) is theoretically ambiguous.

In our shifted-power labor supply example, without screening, the firm hires all workers with $b_i \leq w$, so

$$L_2 = F(w_2) = (w_2 - \mu)^\kappa, \quad e(w_2) = \frac{F(w_2 - \phi)}{F(w_2)} = \left(\frac{w_2 - \phi - \mu}{w_2 - \mu} \right)^\kappa.$$

Effective labor is again just the mass of workers who exert effort:

$$E_2 = e(w_2)L_2 = F(w_2 - \phi) = (w_2 - \phi - \mu)^\kappa.$$

Profits are

$$\begin{aligned} \Pi_2(w) &= G(E_2(w)) - wL_2(w) = a \ln \left((w - \phi - \mu)^\kappa \right) - w(w - \mu)^\kappa \\ &= a\kappa \ln(w - \phi - \mu) - w(w - \mu)^\kappa. \end{aligned}$$

The first-order condition $d\Pi_2/dw = 0$ is

$$\frac{a\kappa}{(w_2 - \phi - \mu)} - (w_2 - \mu)^\kappa - \kappa w_2 (w_2 - \mu)^{\kappa-1} = 0,$$

or equivalently

$$\frac{a\kappa}{(w_2 - \phi - \mu)} = (w_2 - \mu)^{\kappa-1} [(w_2 - \mu) + \kappa w_2].$$

This equation implicitly defines the Case-2 efficiency wage w_2 as a function of (a, μ, κ, ϕ) . We show this case in Figure 4b, where we keep all the parameters as in Figure 4a.

Case 3: No screening, and labor demand binds. The labor supply constraint need not bind when effort (hence productivity per worker) rises with the wage. If the efficiency-wage choice of w (i.e., the choice of w ignoring $L \leq F(w)$) yields excess labor, the constraint is slack and the unconstrained solution is optimal.

Because the firm cannot condition hiring on b_i , the average effective labor per hired worker is $e(w_3) = F(w_3 - \phi)/F(w_3)$; this is the share of workers who do not shirk, among workers who are willing to accept the offer. The amount of effective labor employed at the firm is $e(w_3)L_3$. The firm's unconstrained problem picks w_3 and L_3 to solve:

$$\max_{w_3, L_3} G(e(w_3)L_3) - w_3 L_3.$$

with first-order conditions:

$$G'(eL_3) e'(w_3) L_3 - L_3 = 0, \quad G'(eL_3) e(w_3) - w_3 = 0.$$

Using w_3 to denote the firm's profit-maximizing wage, and combining the

two first-order conditions gives the Solow condition

$$\frac{e'(w_3)}{e(w_3)} = \frac{1}{w_3},$$

which, using $e(w_3) = F(w_3 - \phi)/F(w_3)$ and defining $\eta(x) \equiv xf(x)/F(x)$, is

$$\eta(w_3 - \phi) \frac{w_3}{w_3 - \phi} - \eta(w_3) = 1, \quad (\text{A.23})$$

which gives us a solution:

$$w_3 = \frac{\phi(\eta(w_3) + 1)}{1 - \eta(w_3 - \phi) + \eta(w_3)}$$

The wage w_3 is pinned down by the Solow condition and depends only on the outside-option distribution (via η) and ϕ , not on $G(\cdot)$. Given this “Solow wage,” w_3 , employment satisfies

$$G'(e(w_3)L_3) e(w_3) = w_3,$$

so productivity shifts (G') move L_3 at a fixed w_3 (quantity adjustment).

Even without a monopsony markdown, the Solow wage depends on F through $\eta(\cdot)$. Intuitively, raising w_3 increases the mass of non-shirkers $F(w_3 - \phi)$ and, with random rationing, the share of non-shirkers among those willing to work ($F(w)$); both enter $e(w_3)$. Hence the shape of F disciplines the optimal wage via the Solow condition. Note that under constant elasticity η , the Solow wage $w_3 = (1 + \eta)\phi$ is rising in η : a more competitive labor market yields a higher wage even though there is no monopsonistic markdown.

With log production and shifted-power labor supply, but without a labor supply constraint, the firm solves

$$\max_{w_3, L_3} G(e(w_3)L_3) - w_3 L_3 = \max_{w_3, L_3} a \ln(e(w_3)L_3) - w_3 L_3, \quad e(w_3) = \left(\frac{w_3 - \phi - \mu}{w_3 - \mu} \right)^\kappa,$$

subject to $L_3 \leq F(w_3)$, but with the constraint slack at the optimum.

The first-order conditions are

$$G'(eL_3) e'(w_3) L_3 - L_3 = 0, \quad G'(eL_3) e(w_3) - w_3 = 0.$$

Combining them yields the Solow condition

$$\frac{e'(w_3)}{e(w_3)} w_3 = 1.$$

Since

$$\ln e(w) = \kappa [\ln(w - \phi - \mu) - \ln(w - \mu)],$$

we have

$$\frac{d \ln e(w)}{dw} = \kappa \left(\frac{1}{w - \phi - \mu} - \frac{1}{w - \mu} \right) = \frac{\kappa \phi}{(w - \phi - \mu)(w - \mu)}.$$

Thus the Solow condition becomes

$$\frac{\kappa \phi}{(w_3 - \phi - \mu)(w_3 - \mu)} w_3 = 1 \quad \Longleftrightarrow \quad \kappa \phi w_3 = (w_3 - \phi - \mu)(w_3 - \mu).$$

The relevant (positive root) solution is

$$w_3(\phi) = \mu + \frac{(1 + \kappa)\phi + \sqrt{(1 + \kappa)^2 \phi^2 + 4\kappa\phi\mu}}{2}, \quad (\text{A.24})$$

The first-order condition with respect to L_3 is

$$G'(e(w_3)L_3) e(w_3) = w_3 \quad \implies \quad \frac{a}{e(w_3)L_3} e(w_3) = \frac{a}{L_3} = w_3,$$

so

$$\boxed{L_3 = \frac{a}{w_3(\phi)}, \quad e(w_3)L_3 = \left(\frac{w_3 - \phi - \mu}{w_3 - \mu} \right)^\kappa \frac{a}{w_3}.}$$

This is the case illustrated in Figure 4c with the same parameters as Figure 4b, except that ϕ is increased to ensure that the equilibrium is in the labor-demand constrained region.

Shapiro-Stiglitz Limit. Further observe from Equation A.24 that as κ goes to 0, the solution will be a wage equal to the Shapiro-Stiglitz wage $\mu + \phi$, and will be demand constrained so long as the total amount of labor (normalized to 1) is sufficiently large. That is, the condition for involuntary unemployment is $MRPL(L_3) = \mu + \phi > a = MRPL(1)$.

Constant Elasticity Case. Suppose $\mu = 0$, so $\eta = \kappa$ is constant. Then Equation A.24 reduces to $(1 + \eta)\phi$. This case makes it easy to see that as the elasticity of labor supply becomes larger, wages increase, despite labor supply not binding.

A.4 Transitions From Case 3 to Case 2

When firms cannot condition hiring on μ , they may be constrained by either labor demand (Case 3) or labor supply (Case 2). We next consider how a firm may transition between the two cases.

The supply constraint $L \leq F(w)$ is slack iff $L_3 \leq F(w_3)$, i.e.

$$\frac{a}{w_3(\phi)} \leq [w_3(\phi) - \mu]^\kappa.$$

At the boundary $L_3 = F(w_3)$ we have

$$a^*(\phi) = w_3(\phi)[w_3(\phi) - \mu]^\kappa,$$

so for a given efficiency cost ϕ the global optimum is

$$\begin{cases} \text{Case 3 (demand-constrained Solow solution),} & a \leq a^*(\phi), \\ \text{Case 2 (supply-constrained interior solution),} & a > a^*(\phi). \end{cases}$$

Equivalently, for given (a, μ, κ) there exists a cutoff ϕ^* (defined implicitly by $a = w_3(\phi^*)[w_3(\phi^*) - \mu]^\kappa$) such that the global optimum switches from the supply-constrained Case 2 for $\phi < \phi^*$ to the demand-constrained Solow solution (Case 3) for $\phi > \phi^*$.

$$\begin{cases} \text{Case 3 (demand-constrained Solow solution),} & \phi \geq \phi^*(a), \\ \text{Case 2 (supply-constrained interior solution),} & \phi < \phi^*(a). \end{cases}$$

The global optimum switches from the supply-constrained interior solution (Case 2) for $\phi < \phi^*$ to the demand-constrained Solow solution (Case 3) for $\phi > \phi^*$. We show this transition in the assumed parameter values in Figure A.1, along with wages and employment.

We can also show how the global optimum switches from Case 3 to Case 2 as labor demand increases, by varying a . Figure A.2 shows this transition occurs as labor demand increases, with the transition happening from a Case 3 fixed-wage, no-rent-sharing equilibrium with involuntary unemployment to a Case 2 rent-sharing equilibrium with no involuntary unemployment (but equilibrium shirking).

Remarks. These three cases are based on polar assumptions of full or no screening, and are meant to be a first pass at studying how to combine labor discipline and monopsony. More generally, if firms have some (partial) ability to screen, the logic of the full screening case—where firms are labor hungry and yet turn away some workers, and firms' markdown is limited by

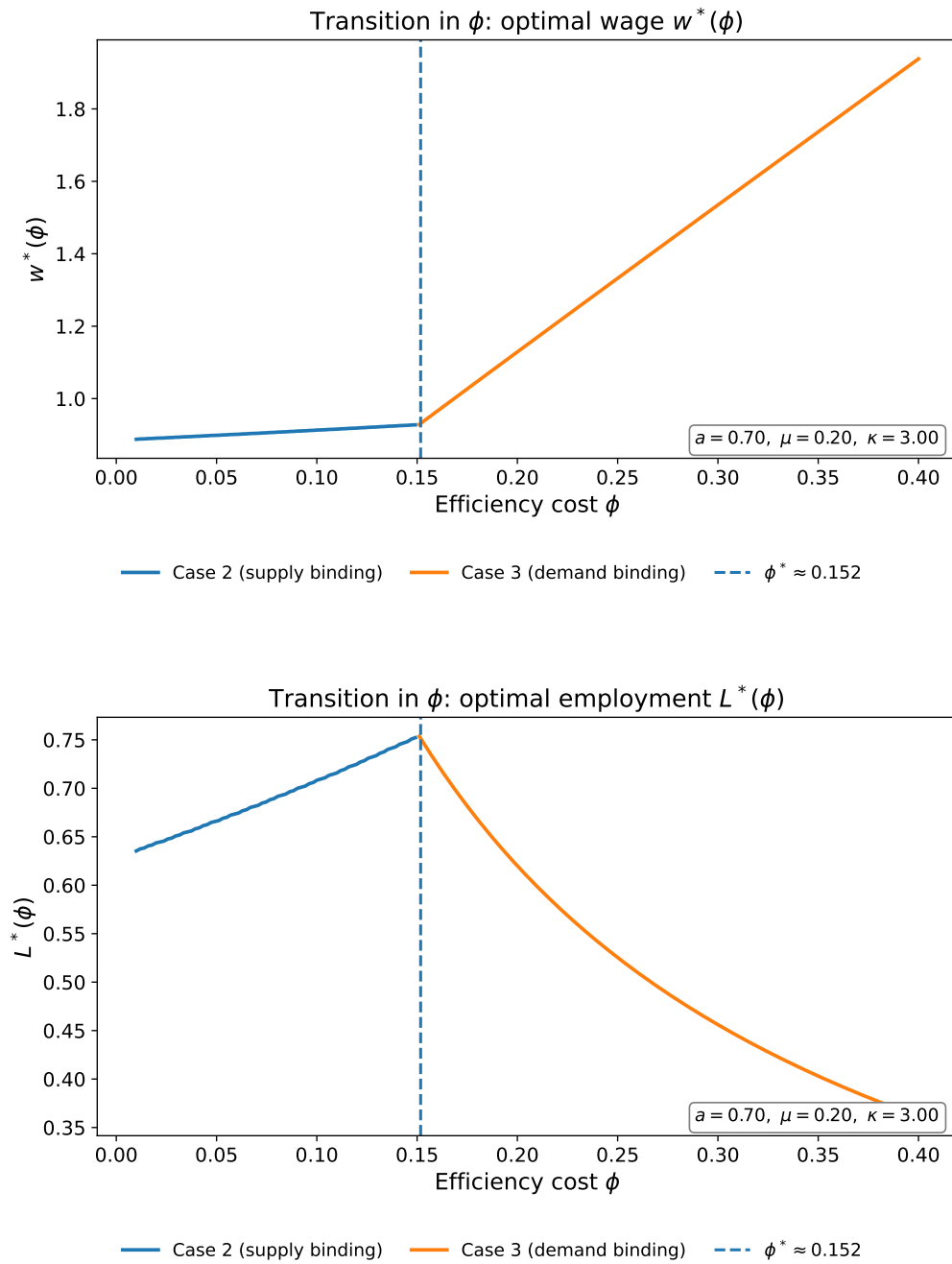


Figure A.1: Employment and wages without screening for values of ϕ , showing transition from labor supply constrained case to labor demand constrained case

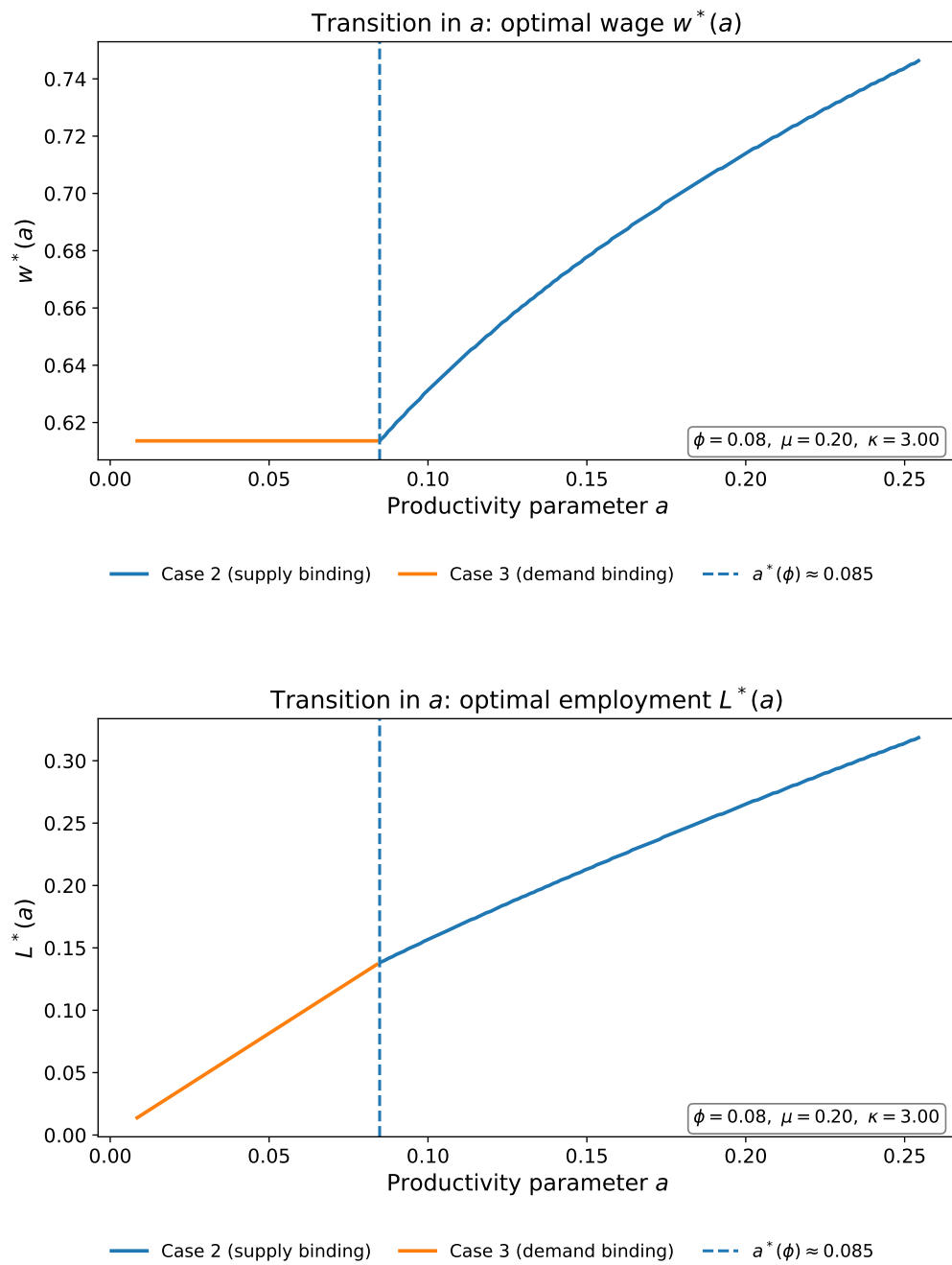


Figure A.2: Employment and wages without screening for values of a , showing transition from labor supply constrained case to labor-demand constrained case.

efficiency wage considerations—may be relevant for a broad segment of the labor market. Analyzing that possibility is a useful direction for future work.

B. Empirical Appendix

B.1 Studies Included in the Forest Plot

Study	Classification	Estimate	Standard Error
Bachmann and Frings (2017)	Calibrated model	2.022	0.160
Bachmann, Demir and Frings (2021)	Calibrated model	1.696	0.030
Dube, Manning, Naidu (2020)	Calibrated model	1.337	0.460
Ransom (2022)	Calibrated model	1.730	0.820
Serrato and Zidar (2016)	Calibrated model	4.188	4.795
Bo, Finan, and Rossi (2013)	Experimental	2.134	0.277
Caldwell and Oehlsen (2023)	Experimental	2.736	2.647
Doerenberg, Duncan, and Löffler (2023)	Experimental	0.440	0.380
Dube, Jacobs, Naidu, and Suri (2020)	Experimental	0.140	0.020
Dube, Naidu, and Reich (2022)	Experimental	4.572	1.456
Muralidharan, Niehaus, and Sukhtankar (2023)	Experimental	3.070	6.960
Portner and Hassairi (2018)	Experimental	0.620	0.080
Amior and Manning (2020)	Non-Experimental	1.007	0.361
Autor, Dube, and McGrew (2024)	Non-Experimental	1.358	0.271
Bassier, Manning, and Petrongolo (2025)	Non-Experimental	0.738	0.004
Bodah, Burkett, Lardaro (2003)	Non-Experimental	0.420	0.111
Booth and Katic (2011)	Non-Experimental	0.709	0.122
Brooks, Kaboski, Li, and Qian (2021)	Non-Experimental	1.175	0.052
Campbell (1993)	Non-Experimental	1.590	0.710
Currie (1991)	Non-Experimental	3.650	1.730
Depew and Sorensen (2013)	Non-Experimental	2.520	0.540
Depew, Norlander, and Sorensen (2017)	Non-Experimental	1.744	0.502
Fleisher and Wang (2004)	Non-Experimental	0.245	0.046
Galizzi (2001)	Non-Experimental	1.820	0.040
Hirsch and Jahn (2015)	Non-Experimental	1.360	0.035
Hirsch, Jahn, Manning, and Oberfichtner (2019)	Non-Experimental	2.220	0.039
Hirsch, Jahn, and Schnabel (2018)	Non-Experimental	2.612	0.289
Hirsch, Schank, and Schnabel (2010)	Non-Experimental	2.489	0.050
Hotchkiss and Quispe-Agnoli (2012)	Non-Experimental	2.370	1.850
Howes (2005)	Non-Experimental	0.680	0.108
Manning (2003)	Non-Experimental	0.547	0.107
Marinescu and Wolthoff (2020)	Non-Experimental	1.620	0.890
Meitzen (1986)	Non-Experimental	1.207	0.673
Ogloblin and Brock (2005)	Non-Experimental	-0.313	0.008
Roussille and Scuderi (2023)	Non-Experimental	4.130	1.130
Sullivan (1989)	Non-Experimental	1.270	0.210
Vick (2017)	Non-Experimental	1.630	0.020
Wasylenko (1977)	Non-Experimental	4.505	1.522
Dobbelaere and Mairesse (2013)	Production Function	2.574	1.099
Felix (forthcoming)	Production Function	1.020	0.008
Pham (2023)	Production Function	1.570	0.965
Tortarolo and Zarate (2020)	Production Function	7.620	0.043
Townsend and Allen (2024)	Production Function	3.446	0.544
Yeh, Macaluso, and Hershbein (2022)	Production Function	1.880	2.520
Amodio and Roux (2023)	Quasi-experimental demand shock	2.439	1.203
Bloesch, Larsen, and Yding (2024)	Quasi-experimental demand shock	1.490	0.335
Cho (2018)	Quasi-experimental demand shock	4.808	2.422
Dhyne et al. (2024)	Quasi-experimental demand shock	3.865	2.387
Goolsbee and Syverson (2023)	Quasi-experimental demand shock	5.051	2.066
Kline, Petkova, Williams, and Zidar (2019)	Quasi-experimental demand shock	3.240	1.700
Kroft, Luo, Mogstad, and Setzler (2020)	Quasi-experimental demand shock	3.500	1.120
Lamadon, Mogstad, and Setzler (2022)	Quasi-experimental demand shock	6.520	0.560

Volpe (2024)	Quasi-experimental demand shock	6.350	0.595
Arnold (2021)	Quasi-experimental wage-shock	0.870	0.440
Barth and Dale-Olsen (2009)	Quasi-experimental wage-shock	1.310	0.043
Bassier, Dube, and Naidu (2022)	Quasi-experimental wage-shock	4.200	0.054
Blau and Kahn (1981)	Quasi-experimental wage-shock	2.356	1.075
Datta (2023)	Quasi-experimental wage-shock	5.635	2.028
Derenoncourt and Weil (2025)	Quasi-experimental wage-shock	2.040	0.420
Dube, Giuliano, and Leonard (2019)	Quasi-experimental wage-shock	4.600	9.000
Falch (2010)	Quasi-experimental wage-shock	1.449	0.387
Falch (2011)	Quasi-experimental wage-shock	7.000	2.293
Falch (2017)	Quasi-experimental wage-shock	1.390	0.200
Gottlieb et al. (2025)	Quasi-experimental wage-shock	0.437	0.109
Matsudaira (2014)	Quasi-experimental wage-shock	-4.384	2.977
Naidu, Nyarko, and Wang; (2016)	Quasi-experimental wage-shock	1.052	0.292
Ransom and Simms (2010)	Quasi-experimental wage-shock	3.691	1.162
Staiger, Spetz, and Phibbs (2010)	Quasi-experimental wage-shock	0.127	0.185
Sulis (2011)	Quasi-experimental wage-shock	0.859	0.012

Note: We define outlier observations as having a labor supply elasticity of more than 15 or less than -10 or a standard error greater than 15.

B.2 Outliers

Study	Classification	Estimate	Standard Error
Fakhfakh and Fitzroy (2005)	Non-Experimental	18.182	6.667
Dobbelare and Vancauteran (2017)	Production Function	11.427	50.400
Tucker (2018)	Quasi-experimental demand shock	15.550	5.150
Mas (2017)	Quasi-experimental wage-shock	18.347	8.629

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